

CYBORG NEWS

NEWSPAPER.CYBORGLAB.ORG

SUNDAY, SEPTEMBER 27, 2026 · VOL. II

FREE TO HUMANS

BONSAI MODEL - 2-billion parameters local	BONSAI QUANT - 1-bit smart glasses run	MODEL SHRINK ▲ 4x keeps performance	CHARTREVIVE - 5 models 5 chart types	RUPTURE LLMS - 3 LLMs 21 conversations	STUDENTBENCH ▲ 175,000 student messages	TUTORING TRIAL - 2,383 human participants
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What's *News*

In brief

- Phone photos leak traits** Researchers built Priva-See to show app access to photos and calendars enables LLM inference of sensitive user traits.
- Charts rebuilt by MLLMs** Benchmarking five MLLMs across five chart types found GPT-5.4 best for ChartRevive reconstruction with verification.
- Social simulation evolves** SocioVerse2 adds a longitudinal simulation loop and controllable research loop for interventions and counterfactuals.
- VR hallucination cues tested** A 24-person VR study found embodied, icon and text cues all helped users spot hallucinations with different trust costs.
- Socratic TA wins** A 132-student randomized trial rated the Socratic full-context AI teaching assistant highest for programming help.

Also noted

- Deregulation harms security** A scholar argues Trump's anti-regulation push to beat China undermines U.S. national security on AI.
- LLMs miss rupture repair** Across 21 conversations judged by 22 experts, LLMs spotted explicit ruptures but offered only moderately effective scripted repairs.
- Kids define trustworthy bots** Among 115 learners who built 119 chatbots, younger children equated trust with obedience while older linked it to transparency.
- MIT interns in Tulsa** MIT students interned with Muscogee health, Cherokee council and a science camp blending AI engineering and philosophy.
- Judgment-centered teaching urged** A 2023-2026 review finds GenAI aids explanations and completion but learning hinges on scaffolding and verification.

DEFENSE AI

D.C. Circuit Lets Pentagon Blacklist Anthropic Claude

A 2-1 majority says refusal to enable military functions meets the procurement law definition of supply-chain risk.



BLACKLISTED The exterior facade of the Pentagon on a clear day. The court left in place the Defense Department ban on Anthropic technology.
JACK BOUCHER · PUBLIC DOMAIN

The U.S. Court of Appeals for the District of Columbia Circuit upheld the Pentagon blacklist of Anthropic in a 2-1 decision. The court found the Defense Department could designate Claude a supply-chain risk under federal procurement law after the company refused to permit military uses. The ruling leaves the ban in place while a conflicting California decision remains under appeal.

The majority said the law covers denial of function, not only sabotage by adversaries. It held that Anthropic disabling

features sought by the Defense Department fit the definition in section 4713. The opinion stated “case raises profoundly difficult questions about the appropriate military uses of an almost unimaginably powerful new technology.” The court added Trump and Defense Secretary Pete Hegseth “must determine how best to balance the competing risks,” and found no breach of authority.

The dissent by Judge Karen Henderson said the statute targets infiltration of federal systems by hostile actors, not open enforcement of use limits. A

judge in the Northern District of California had rejected the designation under section 3252, which requires bad motive by an adversary. The appeals court said it had exclusive authority under the broader procurement provision. It left that California order intact while upholding the second designation.

“As Anthropic admits, the company encodes restrictions into Claude that prevent the model from performing tasks that Anthropic wishes to prevent.”

MAJORITY OPINION, D.C. CIRCUIT

Anthropic sued in March after federal agencies were ordered to stop using its products and defense contractors were barred from business with the company.

MENTAL HEALTH

MIT tool flags suicide risk in crisis texts

The system searches counselor conversations for phrases tied to 49 established risk factors.

MIT scientists built a language-processing tool that estimates suicide risk from text messages exchanged with crisis counselors. The system searches messages for phrases linked to 49 established risk factors and assigns weights to predict severity. Researchers report that the tool accurately predicted risk in new conversations it had not seen.

The tool was developed by Daniel Low, a former graduate student in Senior Research Scientist Satra Ghosh's Senseable Intelligence Group who is now a research scientist at the Child Mind Institute. The team built a lexicon with about 60 words or phrases for each risk factor, with relevance confirmed by expert clinicians. They

trained a machine learning model on de-identified texts from approximately 16,000 conversations with Crisis Text Line volunteer counselors.

Crisis Text Line assessments grouped the conversations into three risk levels: non-suicidal, suicidal ideation without imminent risk, and imminent risk. The imminent risk group included people with a plan for suicide or intent to die within the next 48 hours. “We wanted to know what type of symptoms predict the highest suicide risk,” Low says. The researchers sought patterns visible during a crisis rather than recalled afterward in surveys.

“We wanted to know what type of symptoms predict the highest suicide risk,”

DANIEL LOW, CHILD MIND INSTITUTE

The model found that mentions of lethal means and substance use were more likely to be expressed by the highest-risk group than depressed mood or fatigue. Expressions of active suicidal ideation and self-injury were also strong predictors. Intermediate predictors included anxiety, post-traumatic stress disorder, and emotional pain. The model assigns a weight to each factor based on its contribution to risk.

PHILANTHROPY

\$10 Million Funds 50 Psychiatric Research Fellowships

Five two-year awards for graduate students and postdocs will be made each year for a decade.

Patricia and James Poitras have committed \$10 million to create 50 two-year fellowships for MIT graduate students and postdocs studying major mental illness. The program will award five fellowships each year for the next decade through the Poitras Center for Psychiatric Disorders Research at the McGovern Institute for Brain Research. The gift targets early career scientists focused specifically on psychiatric disorders.

The \$10 million gift addresses severe depression and anxiety, bipolar disorder, schizophrenia and other complex psychiatric conditions. It extends Poitras family support for mental health research at MIT to over \$100 million. The investment focuses on PhD students and postdocs

LEARNING

AI tutoring matches human tutoring on GRE

StudentBench trial of 2,383 learners reports equivalent gains at a fraction of cost.

StudentBench compared AI tutoring with expert human tutoring across 2,383 participants answering Quantitative and Verbal GRE questions. AI tutoring was statistically equivalent to expert human tutoring for GRE learning gains. The comparison used more than 175,000 student-AI messages to study whether large language models produce learning gains equivalent to human tutoring. Participants received AI tutoring, human tutoring, or no tutoring.

The best performing AI tutor surpassed the human tutor on average in five of the seven GRE domains. Statistical equivalence for overall GRE learning gains reached p equals .015. A second study asked expert human tutors to compare lesson plans and practice problems generated by large language models. The comparison used 2,028 pairwise rubric evaluations. Together the two studies separate AI tutors across lesson planning, practice-problem creation, conversational pedagogy, cost and engagement.

One AI tutor achieved learning gains equivalent to human tutoring at 918 times lower cost. Equivalence in that comparison reached p equals .044.

NEUROSCIENCE

Small brain zone sustains fear after amygdala burst

Researchers link continuous defensive behavior in mice to Drd2-positive neurons in the amygdalostratial transition zone.

A Neuron study identifies the amygdalostratial transition zone as the structure that sustains defensive behavior after brief threat signals from the amygdala fade. The work was led by Fergil Mills at the University of Utah with Kay M. Tye at the Salk Institute for Biological Studies. Experiments in mice show neurons in the small zone fire for the full duration of a threat cue. The finding bridges learning about danger and maintaining a defensive state.

The question centers on timing. A 1995 study of rats showed fear learning neurons in the amygdala respond to threat sounds in a brief rapid burst lasting only milliseconds. A 2002 laboratory study showed warning signals travel rapidly from the amygdala directly into the neighboring zone. A 2009 study of rodents described the puzzle that amygdala signals fade almost immediately even though defensive behaviors persist for tens of seconds.

The team first examined genetic identity. They extracted and analyzed the genetic material of 97,434 individual cell nuclei from the brains of mice. The analysis indicated the zone has a distinct genetic identity. It contains an unusually high concentration of Drd2 positive neurons which carry a receptor for dopamine. About 71 percent of the neurons were of this type compared to just 26 percent for Drd1a positive neurons.

“The neural and behavioral results that we observed were striking,”

FERGIL MILLS, UNIVERSITY OF UTAH

The team then measured electrical activity in 15 live mice. The mice were trained where a 20 second auditory tone predicted a mild foot shock while a different 20 second tone predicted a sweet liquid reward. Neurons in the zone responded strongly to the shock predicting tone.

Unlike brief bursts in the amygdala the neurons fired continuously for the entire 20 second duration of the threat cue. Trials with the highest freezing showed firing rates over three times higher than trials with low defensive behavior.

The researchers tested causation with optogenetics which allows control of specific neurons using light. They tested 18 mice engineered so neurons in the zone could be activated by blue light lasers. Light delivery caused the mice to freeze immediately. In a separate test where one side of an enclosure triggered the laser the mice avoided that side. Tests in 35 additional mice isolated cell types. Activating only Drd2 positive neurons produced freezing and avoidance while activating Drd1a positive neurons had no effect on these behaviors. A rotating rod test showed freezing was defensive not a general inability to move.

Calcium imaging tracked the cell types during the tone task using miniature microscopes attached to the heads of 15 mice. Calcium levels rise when neurons

are active. Drd2 positive neurons maintained high sustained activity throughout the threat predicting tones. The team then silenced Drd2 positive neurons in 21 mice with a different optogenetic tool. Blocking these neurons reduced defensive behaviors such as freezing or sudden darting by nearly 50 percent. Blocking did not change responses to reward cues nor general movement in an open arena.

The experiments used a single foot shock threat and a single sweet drink reward. The zone responded to specific warning tones not fear associated contexts. The authors view the circuit as complementary to amygdala learning circuits.

CHARTS

Recovering both the underlying data and visual design from chart images remains a labor-intensive manual process.

COUNSELORS

A new language-processing tool accurately predicts suicide risk from text conversations with crisis counselors.

PLATFORM REGULATION

EFF says EU Kids Act mandates intrusive age checks

The proposal would phase youth access by age and make safety the default, with verification for adults.



UNDER REVIEW Teenagers look at smartphones together on a park bench. Under the draft, such routine use would sit behind age gates and parental approvals.

GAUTHIER DELECROIX - 从青岛, 中国 - CC BY 2.0

The Electronic Frontier Foundation says the EU Commission draft EU Kids Act would place online services behind age gates and require intrusive verification. The group says the measure would expand checks across mainstream platforms and reduce privacy for adults as well as minors. The proposal sets phased access rules for social media and video sharing.

The draft creates three tiers. There are no service accounts for children under 13. Accounts for users from 13 to 15 require tight parental supervision. Users from 15 to 18 receive autonomous accounts in a safe by design environment. Full access is reserved for adults. Parents must set up accounts and prove

parental status. Checks rely on the EU age verification scheme. Providers may skip verification for existing accounts only with a high degree of confidence the user is above the threshold.

The second pillar is safety by design. It covers social media, video sharing, online games, AI companions, chatbots and app stores. Providers must make child safe design the default and may relax rules only after age assurance shows the user is an adult. Rules address infinite scrolling, account settings and choice over recommender systems. AI services must reduce exposure to emotional dependency and harmful interaction. Games must include contact protections.

SPORT TECH

Umpires shift zone toward automated system in 2026

Analysis of 4,114,256 pitches and 8,447 challenges finds temporary adjustment after overruns and many overturnable calls left unchallenged.

The effective strike zone called by Major League Baseball umpires moved toward the automated zone in 2026. The finding comes from 4,114,256 called pitches from 2015 through 2026 and 8,447 challenges from the 2026 season. Umpires make every ball and strike call under the Automated Ball-Strike challenge system. Players can ask the automated system to publicly overturn those decisions. The shift exceeded the trajectory observed in prior seasons. Consistency largely continued its existing trend.

The analysis tracks where umpires placed the effective strike zone boundary and how consistently they

applied that boundary. The work studies umpire judgment and player behavior under algorithmic review. The setting is distinctive. Following an overturned call, umpires temporarily adjusted subsequent decisions near the corrected boundary. Those effects didn't consistently persist into the next game. The adjustment was local and short lived. The result was correction without a lasting reset.

Count dependent variation in calling remained. Differences associated with player status narrowed. Players left many overturnable calls unchallenged.

EDGE AI

PrismML Shrinks Model for Smart Glasses

2-billion-parameter Bonsai version runs locally on Snapdragon AR1 Gen 1 Platform.

PrismML has created a version of its tiny language models for smart glasses running on Qualcomm Snapdragon chips. The 1-bit Bonsai LLM runs locally on AI smart glasses built on the Snapdragon AR1 Gen 1 Platform. The model was showcased at the Snapdragon Summit. It contains 2-billion parameters tuned for vision and language. It is shrunk by 4x from a larger model. Wearers can ask what they are looking at in real time.

The lab was founded by Caltech researchers and is advised by Ion Stoica of UC Berkeley. Its method shrinks larger models substantially while retaining almost all of their performance on standard

benchmarks. The approach targets open-weight AI that runs on devices. It aims to make better use of the computing power they already have. The startup presents local execution as an alternative to dependence on proprietary AI labs.

The startup pitches this approach as an alternative to depending on the privacy promises of proprietary AI labs and their insatiable need for more compute. Releasing a model for the Qualcomm chip is a step toward that vision. No smart glasses running PrismML have been announced yet. The Summit demonstration focused on local vision and language use on glasses hardware.

MODEL RACE

Meta steers Muse toward glasses after launches

Anthropic released Opus 5.5 90 minutes before OpenAI updated GPT-6, while Muse is reportedly ahead of early ChatGPT numbers.

Anthropic released Opus 5.5. OpenAI followed with GPT-6 updates 90 minutes later. Meta took attention with Muse, its personal agent. The agent is reportedly outpacing ChatGPT early numbers. It is headed to smart glasses and a tiny Tamagotchi-style device next. The order of releases framed the week. A consumer product drew the focus. The gap was narrow. The timing was tight. Model drops came in sequence. Muse sits apart from those drops. It is described as a personal agent. That distinction matters.

On TechCrunch Equity, Kirsten Korosec, Anthony Ha and Sean O'Kane examined Meta consumer

AI strategy. They traced where money is flowing. They assessed what it means for startups building on top of this technology. They asked which AI products might become part of people daily lives. The discussion centered on product distribution. Glasses and small devices change placement. Placement changes use. Startups face a platform question. Build above Muse or compete with it. Money follows access to users. Access follows daily habit. Habit forms around concrete tasks. Tasks decide retention.

They also addressed a16z new school for aspiring founders.

LEARNING STUDIES COMPARED		
SAMPLE SIZES BEHIND RECENT TUTORING AND TRUST STUDIES.		
STUDY	TESTED	COUNT
VR cues	embodied cues and baseline	24 participants
AI teaching assistants	three programming tasks	132 students
StudentBench GRE	AI vs human tutoring	2,383 participants
StudentBench corpus	student-AI messages	over 175,000 messages

MODELS UNDER REVIEW		
WHAT EACH SYSTEM WAS TESTED AGAINST.		
SYSTEM	TASK	SCALE
Bonsai LLM	vision for smart glasses	2-billion parameters
Bonsai compression	shrink of larger model	by 4x
ChartRevive	data and design extraction	five MLLMs
Rupture bench	identify and resolve ruptures	three LLMs
Rupture judges	expert evaluations	22 experts

THE PENTAGON BLACKLIST STANDS
Courts let the Pentagon blacklist a model maker. A US appeals court approved the Department of Defense's blacklisting of Anthropic technology after the company withheld certain AI features from the military. In a 2-1 ruling, the US Court of Appeals for the District of Columbia Circuit said the Trump administration had authority to blacklist Anthropic even if Anthropic had no malicious intent. The department had ample support for its conclusion that the continued integration of Claude into the department's information systems presented a statutorily covered national-security risk.